



Lab 01: Cloud Storage Qwik Start - Console

Google Cloud Associate Cloud Engineer | GSP073

Overview

Cloud Storage allows worldwide storage and retrieval of any amount of data at any time. In this lab the Google Cloud Console was used to create a storage bucket, upload objects, organize files with folders, and make objects publicly accessible.

Objectives

- Create a Cloud Storage bucket
- Upload an object to the bucket
- Share a bucket object publicly
- Create folders and subfolders
- Delete a folder

Key Concepts

Term	Definition
Bucket	The basic container that holds data in Cloud Storage. Names must be globally unique.
Object	Any file stored in a bucket. Object names must be unique only within their bucket.
allUsers	A special IAM member that grants public access to anyone on the internet.
Storage Object Viewer	IAM role that allows read access to objects in a bucket.
Uniform Access	Access control that applies permissions at the bucket level using IAM only.

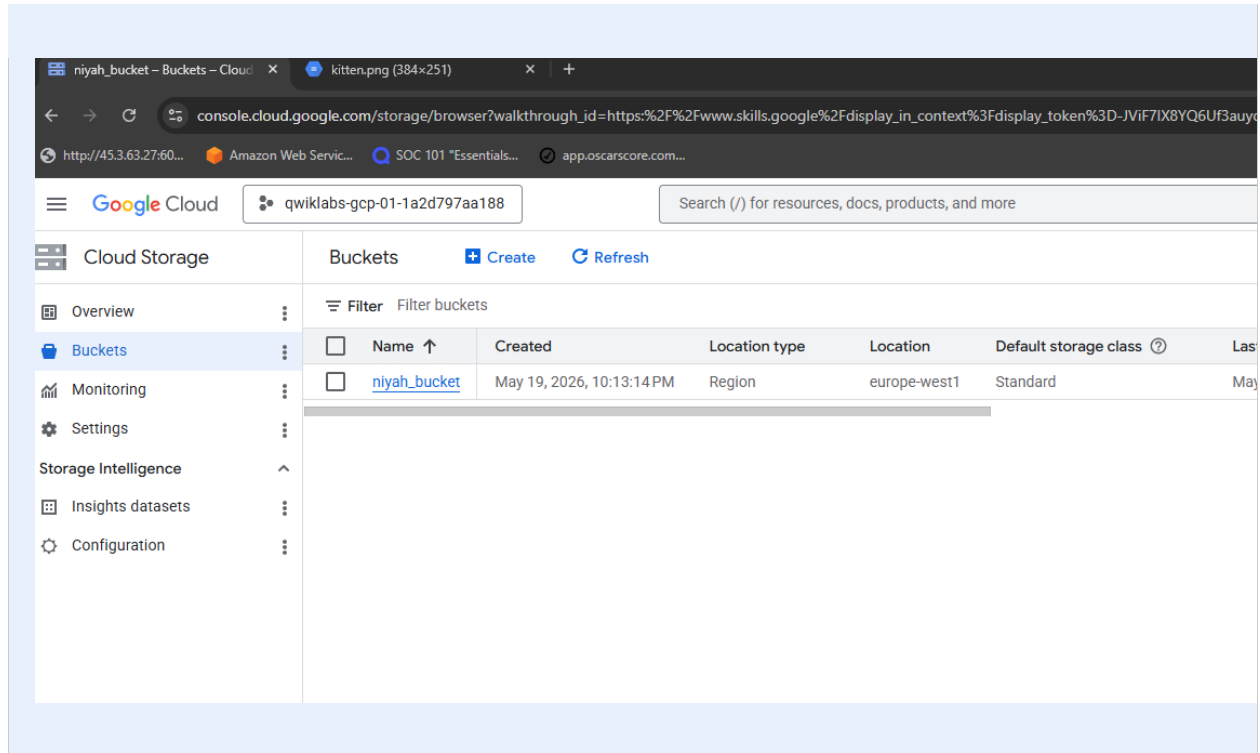
Task 1 — Create a Bucket

Created a new Cloud Storage bucket using the GCP Console with the following settings:

- Name: [niyah_bucket] — must be globally unique
- Location type: Region
- Location: europe-west1

- Storage class: Standard
- Access control: Uniform
- Public access prevention: Off

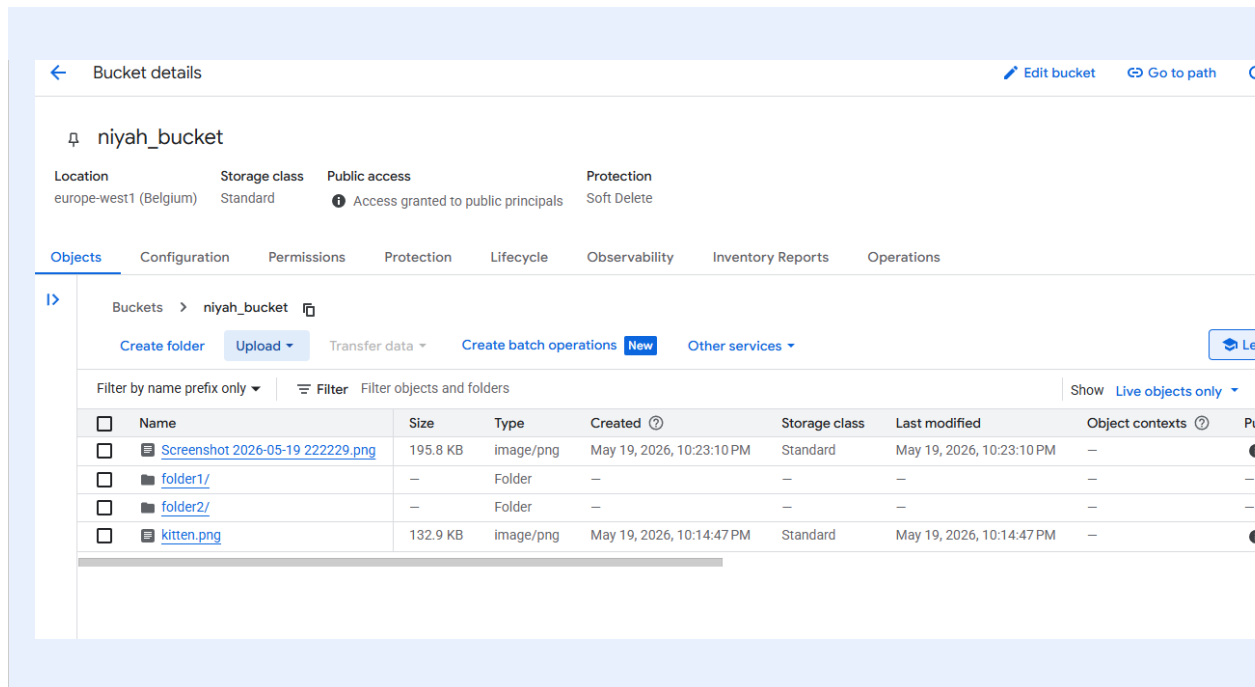
Navigation: Cloud Console → Navigation Menu → Cloud Storage → Buckets → Create



Task 2 — Upload an Object

Downloaded the kitten.png image from the lab and uploaded it to the bucket using the console upload feature.

- Clicked the bucket name to open it
- Clicked Upload → Upload files
- Selected kitten.png from local computer
- Verified file name, size, and type after upload



Task 3 — Share a Bucket Publicly

Made the bucket publicly accessible by granting the allUsers principal the Storage Object Viewer role. This allows anyone on the internet to view objects in the bucket via a public URL.

- Clicked the Permissions tab
- Clicked Grant Access
- New principal: allUsers
- Role: Cloud Storage → Storage Object Viewer
- Clicked Save → Allow public access

Public URL format: `https://storage.googleapis.com/YOUR_BUCKET_NAME/kitten.png`

Google Cloud | qwiklabs-gcp-01-1a2d797aa188 | Search (/) for resources, docs, products, and more

Cloud Storage | Bucket details | Edit bucket | Go to path | Refresh | Learn

Overview | Buckets | Monitoring | Settings | Storage Intelligence | Insights datasets | Configuration

Objects | Configuration | **Permissions** | Protection | Lifecycle | Observability | Inventory Reports | Operations

Public access

Access granted to public principals
Public access may be prevented by an organization policy. You need the `orgpolicy.policy.get` permission to confirm whether this bucket is public.

One or more bucket-level permissions grant access to everyone on the internet (`allUsers`) or anyone signed into a Google account (`allAuthenticatedUsers`). If this bucket should not be publicly accessible, remove these public permissions or prevent public access to this bucket. [Learn more](#)

[Prevent public access](#)

Access control

Uniform: No object-level ACLs enabled
90 days left to change this setting

All object access is controlled by bucket permissions and objects cannot have their own access control lists (ACLs). To allow per-object access, you can switch to fine-grained access within 90 days. [Learn more](#)

[Switch to fine-grained](#)

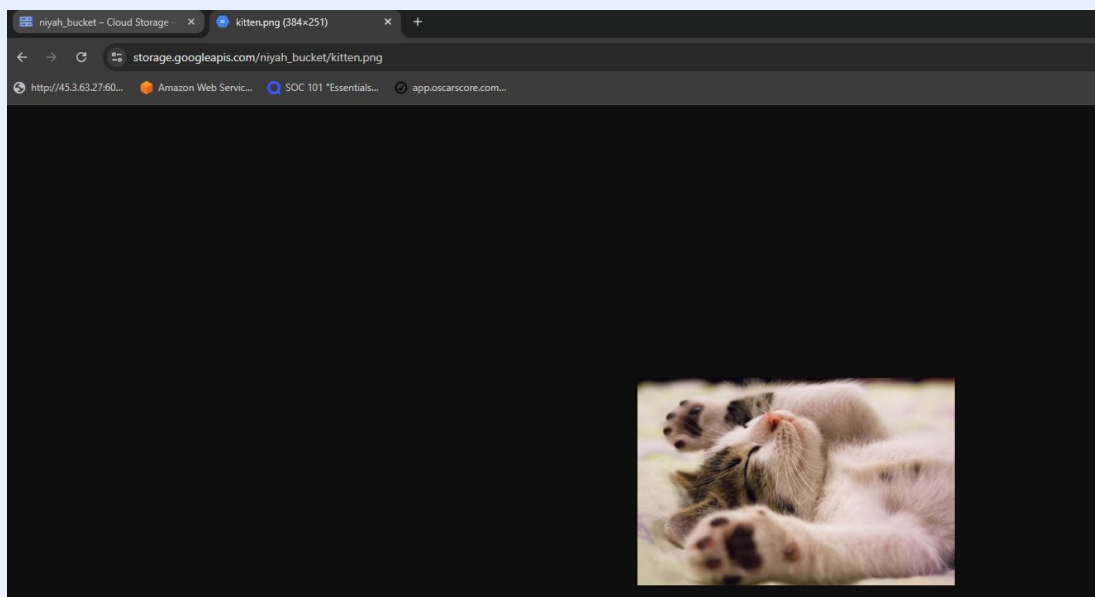
Permissions

[View by principals](#) | [View by roles](#)

[Grant access](#) | [Remove access](#)

Filter Enter property name or value

Type	Principal	Name	Role	Inheritance
☐	653575952552-compute@developer.gserviceaccount.com	Compute Engine default service account	Storage Admin	qwiklabs-gcp-01-1a2d797aa188
☐	allUsers		Storage Object Viewer	
☐	Editors of project: qwiklabs-gcp-01-1a2d797aa188		Storage Legacy Bucket Owner	
☐			Storage Legacy Object Owner	



Task 4 — Create Folders

Created a folder structure inside the bucket and uploaded a file to a subfolder.

- Created folder1 in the bucket root
- Opened folder1 and created folder2 inside it
- Uploaded a file into folder2

← Bucket details

Edit bucket

📁 niyah_bucket

Location

europa-west1 (Belgium)

Storage class

Standard

Public access

🛑 Access granted to public principals

Protection

Soft Delete

Objects

Configuration

Permissions

Protection

Lifecycle

Observability

Inventory Reports

Operations

I>

Buckets > niyah_bucket 📁

Create folder

Upload ▾

Transfer data ▾

Create batch operations

New

Other services ▾

Filter by name prefix only ▾

Filter

Filter objects and folders

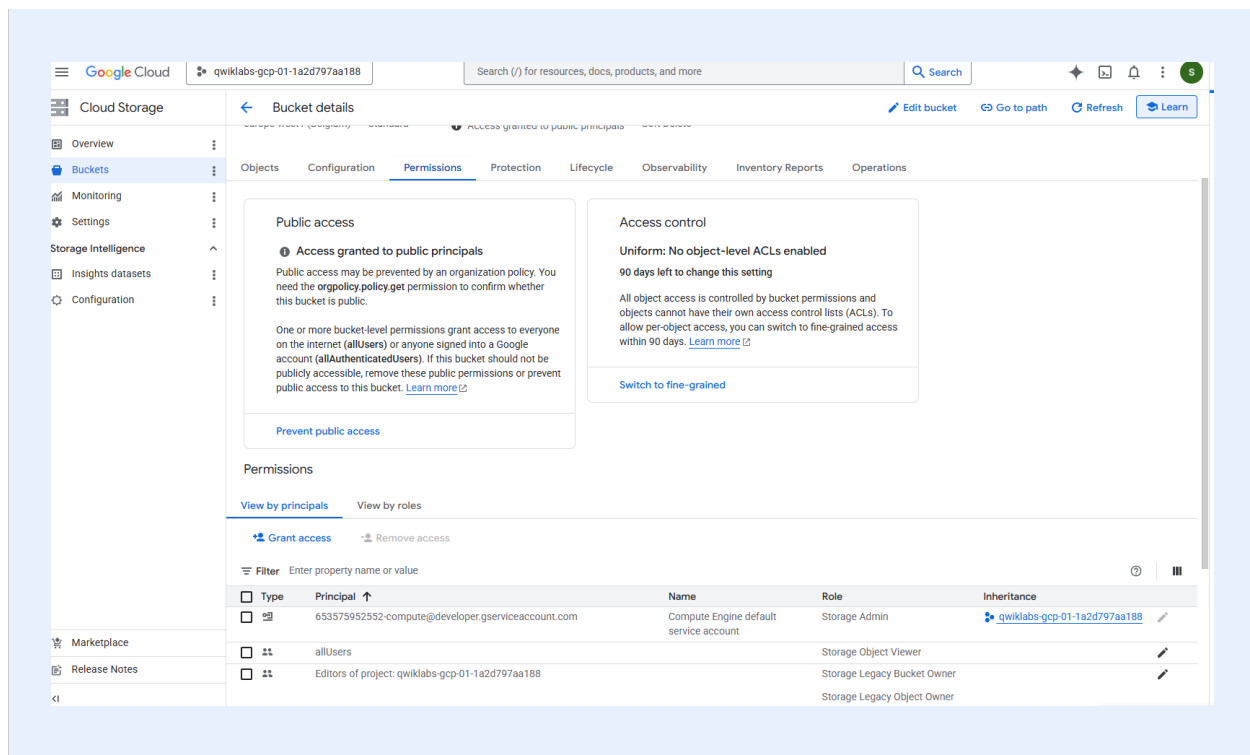
Show

Liv

☐	Name	Size	Type	Created ?	Storage class	Last modified	Object
☐	 Screenshot 2026-05-19 222229.png	195.8 KB	image/png	May 19, 2026, 10:23:10 PM	Standard	May 19, 2026, 10:23:10 PM	—
☐	 folder1/	—	Folder	—	—	—	—
☐	 folder2/	—	Folder	—	—	—	—
☐	 kitten.png	132.9 KB	image/png	May 19, 2026, 10:14:47 PM	Standard	May 19, 2026, 10:14:47 PM	—

Task 5 — Delete a Folder

Deleted the bucket and all its contents by navigating back to the bucket level, selecting it, clicking Delete, and typing DELETE to confirm.



Real World Application

Cloud Storage buckets are used everywhere in production environments:

- Hosting static website assets like images, CSS, and JavaScript files
- Storing backups and disaster recovery data
- Distributing large files like software packages or datasets
- Serving media content like profile pictures or product images in apps

What I Learned

- Bucket names must be globally unique across all of Cloud Storage
- Object names only need to be unique within their own bucket
- Granting allUsers the Storage Object Viewer role makes objects public
- Uniform access control manages permissions at the bucket level via IAM
- Folders in Cloud Storage are virtual — they are part of the object name
- Public objects get a shareable URL at storage.googleapis.com

Challenges & How I Solved Them

Created the bucket without unchecking Enforce public access prevention. This blocked the ability to make objects public in Task 3. Fixed by going back into the bucket Permissions tab and turning off public access prevention before granting

access to allUsers.

Had trouble setting permissions to allUsers because public access prevention was still enabled from the previous mistake. Once it was turned off the allUsers principal was added successfully with the Storage Object Viewer role.

Key lesson: Always read every step carefully during bucket

creation since access settings affect later tasks.

Resources

GCP Cloud Storage Docs: <https://cloud.google.com/storage/docs>
Lab ID: GSP073